

The aim of this project is to make students practice data manipulation using pandas and visualization using matplotlib Python libraries.

- The Project will be conducted in group. The existing group for the previous project will continue.
- Dataset is shared with each group. But the project shall be conducted individually and submitted to the LMS according to the deadline.
- **Duration of the project: 1 week (starting the day it is posted on the LMS)**

Questions

1. Load the dataset using Pandas
2. Display the first 8, 12 and 25 rows separately
3. Check the data type of at least 5 rows separately
4. Check the data type of the whole columns
5. List the columns names of the data frame
6. Copy your data frame to a new data frame and:
 - a. Change the data type of columns whose data type is integer into float
 - b. Change the data type of columns whose data type is float into integer
7. Create a new data frame the consists of columns which are not numeric.
8. Display those whose value is >, >=, <, <=, == a specific column's value (for example, salary>3000)
9. Perform groupings (at least 4 kinds), based on criteria. The grouping shall have aggregation (for example, average, maximum, minimum etc.)
10. Display range of rows (1-5) and first, second, fourth, sixth columns.
11. Display third through tenth rows and the first four columns
12. Perform filtering setting your own criteria based on your dataset
13. Perform slicing using index and column names
14. Perform slicing using index only
15. Check null values for each column separately
16. Display the total number null values (missing values) all columns
17. Identify columns having missed or null values and replace them with:
 - a. Mean of the column
 - b. Minimum value of the column
 - c. maximum value of the column
18. export the filled data frame to new CSV file in the format (**filename_yourname.csv**)
19. using the file in question 18, do histogram, pie chart, bar chart, line graphs and other visualizations using Matplotlib.